n rectangles having $x_i - x_{i-1} = \Delta x_i$ as its base. Then area $A_i = \Delta x_i f\left(x_i'\right)$. Now the total area

$$A = \sum_{i=1}^{n} f\left(x_i'\right)\Delta x_i$$

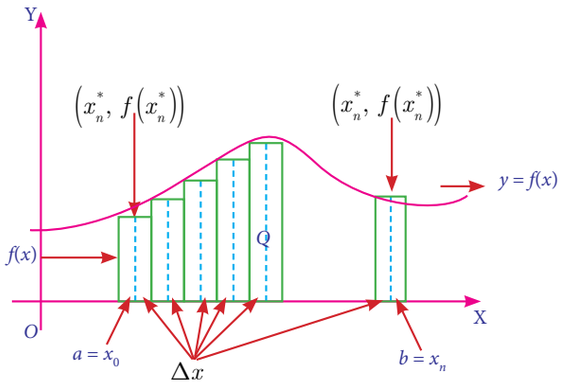


Fig. 3.1

Now from the definition of definite integral, if $f\left(x\right)$ is a function defined on $\left[a,b\right]$ with $a < b$ then the definite integral is

$$\int_a^b f\left(x\right)dx = \lim_{\Delta x_i \to 0} \sum_{i=1}^{n} f\left(x_i'\right)\Delta x_i$$

The area under the curve is exhausted by increasing the number of rectangular strips to ∞

Thus the geometrical interpretation of definite integral is the area under the curve between the given limits.

The area of the region bounded by the curve $y=f(x)$, with x- axis and the ordinates at $x = a$ and $x = b$ given by

$$\text{Area } A = \int_a^b y\,dx$$

$$= \int_a^b f(x)dx$$

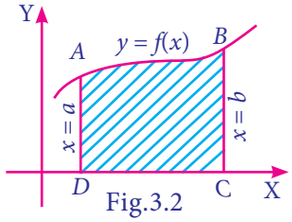


Fig.3.2

Note

(i) The area of the region bounded by the curve $y = f(x)$ between the limits $x = a$, $x = b$ and lies below x -axis, is

$$A = \int_a^b -y\,dx = -\int_a^b f(x)dx$$

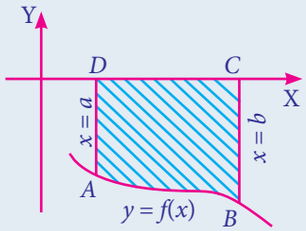


Fig.3.3

(ii) The area of the region bounded by the curve $x = f(y)$ between the limits $y = c$ and $y = d$ *with* $y - axis$ and the area lies lies to the right of y- axis, is

$$A = \int_c^d x\,dy \quad = \int_c^d f(y)dy$$

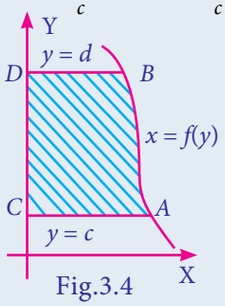


Fig.3.4

(iii) The area bounded by the curve $x = f(y)$ between the limits $y = c$ and $y = d$ *with* $y - axis$ and the area lies to the left of y- axis, is

$$A = \int_c^d -x\,dy = -\int_c^d f(y)dy$$

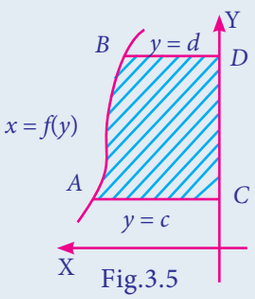


Fig.3.5

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